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Enhancing anti sulfate high-temperature dry wet cycle erosion performance of recycled aggregate concrete:Nano-SiO₂ and fiber synergistically composite alkali activated

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Outline

Highlights:

Abstract

Keywords

1. Introduction

2. Materials and manufacturing

3. Experimental

4. Results and discussion

5. Conclusions

Highlights:

- PPF(0.3%) and RCF(0.5%) enhance RAC strength and durability.
- NS improves C-(A)-S-H formation, reducing porosity in RAC.
- Synergistic NS-fiber effect boosts sulfate resistance in SSR-RAC.

Abstract

The recycling of industrial solid waste in construction engineering promotes sustainable development. However, the performance degradation of fully recycled concrete under coupled high-temperature dry-wet cycles and sulfate attack remains a critical barrier to

its widespread application. This study utilizes an eco-friendly cementitious material, Soil Solidification Rock (SSR), derived entirely from industrial solid waste, as a highly efficient replacement for cement, alongside 100% recycled coarse aggregates substituting natural aggregates, to prepare alkali-activated recycled aggregate concrete (RAC). By integrating nano-SiO₂ (NS) and fiber-reinforced composite alkali-activation technology, the resistance of RAC to sulfate attack under high-temperature dry-wet cycles was investigated. Four mix designs were developed, comparing single NaOH and composite NaOH-Na₂SiO₃ alkali-activation systems, with NS modification. The influence of alkali-activation systems on the setting characteristics of RAC was evaluated. Using NS, polypropylene fibers (PPF), and recycled carbon fibers (RCF) as variables, the deterioration mechanisms of RAC were elucidated through analyses of surface morphology, mass change rate, compressive strength, and characterization techniques including X-ray diffraction (XRD), Fourier Transform Infrared Spectroscopy (FTIR), and Scanning Electron Microscope(SEM). The results demonstrate that NS enhances the compressive strength of single alkali-activated concrete more effectively than the NaOH–Na₂SiO₃ composite system. The incorporation of 0.3% PPF and 0.5% RCF individually increased the compressive strength of RAC by 22.2% and 15.2%, respectively. After 75 days of sulfate exposure under high-temperature dry–wet cycles, the NT12-P20-C30 specimen exhibited a 70.2% compressive strength corrosion–resistance coefficient—slightly below the 75% failure threshold, yet still comparatively superior within this study's material system.The pozzolanic effect of NS facilitated the formation of C-(A)-S-H, optimizing the interfacial transition zone and reducing porosity. Fiber reinforcement enhanced matrix bonding, effectively suppressing microcrack propagation. The synergistic effect of NS and fibers significantly improved the pore structure and interfacial properties of RAC, enhancing its mechanical performance and resistance to deterioration. These findings provide theoretical and technical support for the engineering application of SSR-based fully recycled aggregate concrete.

Keywords

CRedit authorship contribution statement

Declaration of competing interest

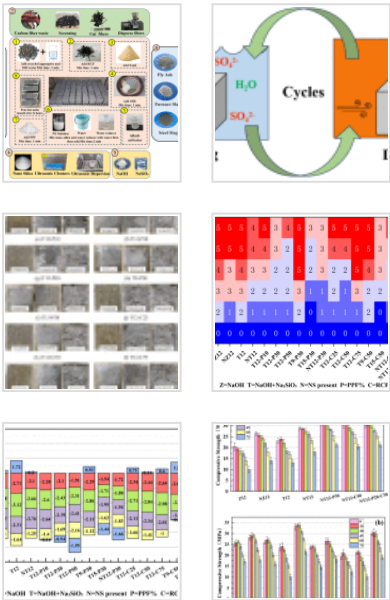
Acknowledgments

Data availability

References

Show full outline

Figures (11)



Show 5 more figures

Tables (5)

Table 1. Oxide composition of gelling ma...

Table 2. Physical indicators of aggregates.

Composite alkali-activation system; Fully recycled aggregate concrete; Nano-SiO₂;
Polypropylene fibers; Recycled carbon fibers; Resistance to sulfate attack

☒ Table 3. Physical properties of nano-silica.

☒ Table 4. Fiber performance specifications.

☒ Table 5. Mixing ratio and material consu...

[< Previous article in this issue](#)

[Next article in this issue >](#)

1. Introduction

With the rapid pace of global urbanization and infrastructure development, the construction industry faces an ever-increasing demand for cement and natural aggregates. However, traditional cement production is energy-intensive and accounts for approximately 8% of global greenhouse gas emissions, exerting significant environmental pressure and posing challenges to the industry's sustainability [[1], [2], [3]]. Concurrently, the overexploitation of natural aggregates has led to resource depletion, land degradation, and ecosystem destruction, underscoring the urgent need for eco-friendly alternative materials to mitigate resource crises and environmental burdens [1,4]. In recent years, the development and application of green, low-carbon building materials have emerged as a critical pathway for achieving sustainable development in the construction sector. Soil Solidification Rock (SSR), a novel cementitious material derived primarily from industrial solid wastes such as fly ash, slag, and steel slag, utilizes alkali-activation technology to produce compounds with excellent binding properties. SSR not only serves as an effective substitute for traditional cement but also significantly reduces carbon footprint and environmental impact [[5], [6], [7]]. Additionally, recycled aggregate concrete (RAC), which employs construction and demolition waste to produce recycled coarse aggregates, replaces natural aggregates, thereby alleviating the accumulation of solid waste and promoting the circular utilization of construction resources. This approach provides a new pathway for the industry's green transformation [[8], [9], [10], [11], [12]]. In this context, the synergistic use of SSR and RAC can help reduce reliance on natural resources

and provide a feasible approach for the circular utilization of construction materials, thereby promoting more sustainable practices within the construction industry.

However, the high porosity, elevated water absorption, and crushing index of recycled aggregates result in inferior mechanical properties and a loose microstructure in the interfacial transition zone (ITZ), particularly under harsh environmental conditions such as sulfate attack [[13], [14], [15]]. These durability limitations significantly hinder the engineering applications of RAC. Xie et al. [3] investigated the mechanical properties and durability of RAC under sulfate attack, revealing that different replacement ratios of recycled aggregates substantially affect concrete durability. Wei et al. [7] demonstrated that the fly ash content significantly influences the performance of Soil Solidification Rock Recycled Aggregate Concrete (SSRRAC). When the fly ash content is 30% or lower, the sulfate resistance of SSRRAC improves with increasing fly ash incorporation. Qi et al. [16] found that the sulfate resistance of RAC exposed to dry-wet cycles is markedly influenced by the replacement ratio of recycled aggregates, with the addition of fly ash (FA) and ground granulated blast-furnace slag (GBFS) enhancing its durability. Benli et al. [17] confirmed that the combined incorporation of fly ash and silica fume improves the sulfate resistance of concrete, with compressive and tensile strengths increasing up to 90 days but declining after 180 days.

In recent years, alkali-activation technology has demonstrated significant potential in enhancing the performance of concrete materials. Ma et al. [18] systematically investigated the effects of different sodium silicate alkali activators on geopolymer concrete, revealing that the type of sodium silicate activator significantly influences reaction products and microstructure. Studies [19] have shown that composite activators, such as calcium hydroxide, sodium sulfate, and sodium carbonate, can effectively enhance the strength and microstructural density of alkali-activated slag concrete (AASC), although long-term use may lead to increased shrinkage and creep. Ganesh et al. [20] noted that the strength of geopolymer concrete is influenced by the binder-to-alkaline solution ratio, the NaOH-to-Na₂SiO₃ ratio, and the molar concentration of NaOH. The type

of alkali activator (e.g., NaOH, Na₂SiO₃, KOH) significantly affects the material's sulfate resistance by regulating the composition of the gel phase and the pore structure.

Fiber reinforcement and nanomaterial modification are effective strategies for enhancing concrete durability. Wang et al. [21] demonstrated that basalt fibers can optimize the pore structure of concrete in sulfate attack environments, significantly improving compactness and slowing pore expansion and deterioration processes. Ren et al. [22] investigated the durability of polypropylene fiber (PPF)-reinforced concrete under coupled freeze-thaw cycles and sulfate attack, finding that PPF improved durability by 67% through a bridging effect. Liang et al. [23] further validated the sulfate resistance of multi-scale PPF-reinforced concrete, showing that after 20 sulfate dry-wet cycles, its compressive strength surpassed that of ordinary concrete. PPF, as a well-established reinforcement material [[24], [25], [26]], significantly enhances the tensile strength and toughness of concrete by inhibiting crack propagation. Meanwhile, recycled carbon fibers (RCF), an emerging eco-friendly reinforcement material, have gained attention in recent years due to their high strength and sustainability [[27], [28], [29]]. Although research on RCF applications in concrete reinforcement is relatively limited, its high modulus and bridging effect offer substantial potential for suppressing crack propagation and enhancing mechanical properties, particularly in harsh environments such as sulfate attack.

Nano-silica (NS), due to its high specific surface area and pozzolanic effect, generates additional C-(A)-S-H gel, filling micropores and optimizing the microstructure of the ITZ [30,31]. Gao et al. [32] found that NS increases the proportion of micropores smaller than 100nm in high-volume slag (HVS) materials while reducing macropores larger than 100nm, thereby optimizing pore distribution. Xu et al. [33] reported that NS promotes the reaction between ground granulated blast-furnace slag (GGBS) and calcium hydroxide (CH), enhancing the reactivity of GGBS during early hydration stages. Studies indicate that NS incorporation effectively reduces the mass change rate of mortar after sulfate attack, minimizes the formation of degradation products, and delays the erosion process [34]. Consequently, the synergistic modification with PPF, RCF, and NS offers new possibilities for enhancing the mechanical properties and durability of concrete.

Existing studies have demonstrated that single modification techniques, such as NS incorporation or fiber reinforcement, can enhance the performance of RAC. However, the synergistic mechanisms of NS and fibers in composite alkali-activated RAC, particularly their impact on durability under sulfate attack coupled with high-temperature dry-wet cycles, remain underexplored. Furthermore, while most research focuses on traditional cement-based RAC or concrete with partial recycled aggregate replacement, the microscopic degradation mechanisms of SSR based concrete with 100% recycled aggregates in extreme environments are not yet fully understood. Therefore, this study aims to leverage the performance-enhancing potential of composite alkali-activation systems, combined with the multi-scale and multi-dimensional reinforcement effects of NS and fibers, to investigate the mechanisms underlying the improved performance of fully recycled concrete in harsh environmental conditions.

To this end, this study employs SSR as the cementitious material (composed of 75% fly ash/steel slag/slag in a 1:1:1 ratio, 15% gypsum, 5% cement, and 5% silica fume) and 100% recycled coarse aggregates as the aggregate. NaOH and NaOH-Na₂SiO₃ composite alkali-activation systems were designed to systematically investigate the effects of synergistic modification with NS, PPF, and recycled carbon fibers RCF on the setting characteristics, mechanical properties, and sulfate resistance of RAC. In alkali-activated systems, the incorporation of a small amount of Portland cement supplies additional Ca²⁺ and promotes the early formation of C-S-H gels, thereby accelerating setting and hardening, refining the gel structure, improving workability, and significantly enhancing both mechanical properties and durability. This represents an effective synergistic activation strategy. Through sulfate attack tests under high-temperature dry-wet cycles, the durability of modified RAC was evaluated using surface morphology, mass change rate, and compressive strength tests. Additionally, X-ray diffraction (XRD), Fourier-transform infrared spectroscopy (FTIR), and scanning electron microscopy (SEM) were employed to analyze microstructural changes before and after erosion, elucidating the degradation mechanisms of SSRAC under the synergistic action of NS and fibers. This study aims to explore the intrinsic mechanisms governing the performance of fiber- and nano-

enhanced alkali-activated RAC under the coupled effects of high-temperature dry-wet cycles and sulfate attack in harsh environments. The findings provide theoretical and technical support for its application in high-risk sulfate environments, such as bridges and ports, while promoting the resource utilization of industrial solid waste and the development of green building materials.

2. Materials and manufacturing

2.1. Materials

2.1.1. Gelling materials

The gelling materials used in the preparation of concrete mainly consist of fly ash, steel slag, ore slag, gypsum, silica fume and cement,. The fly ash used is secondary low-calcium fly ash (CaO <10%). The ore slag is S95 granulated blast furnace slag powder, and the selected steel slag is in the form of black powder. The oxide compositions of each material are determined through X-ray Fluorescence (XRF) spectroscopy analysis of the samples, and the oxide compositions are presented in [Table 1](#).

Table 1. Oxide composition of gelling materials (%).

Form	SiO ₂	Al ₂ O ₃	CaO	Fe ₂ O ₃	K ₂ O	MgO	SO ₃	Other
Fly ash	53	33	5.6	2.5	—	0.95	—	4.95
Ore slag	33.06	15.04	39.29	—	—	9.96	1.9	0.75
Steel slag	19.4	6.37	38.72	22.42	—	6.05	0.81	6.23
Gypsum	6.5	2.18	65.12	1	0.71	2.65	—	21.83
Silica fume	96.2	—	—	—	—	—	—	3.8

Form	SiO ₂	Al ₂ O ₃	CaO	Fe ₂ O ₃	K ₂ O	MgO	SO ₃	Other
cement	17.7	8.90	63.14	4.52	0.94	—	3.74	1.06

2.1.2. Aggregates

The recycled coarse aggregates used in this study were sourced from construction waste generated during the expansion and reconstruction project of the Rong Wu Expressway (Yantai Hub to Penglai Hub section) and processed professionally. The raw materials primarily consisted of concrete, mortar, stone, and bricks. The processed recycled coarse aggregates had a particle size range of 4.75 mm to 31.5 mm, meeting the continuous gradation requirements for 5–31.5 mm aggregates. The particle size distribution was proportioned with a mass ratio of 1:1.5:1.5 for the size ranges of 4.75–9.5 mm, 9.5–19 mm, and 19–31.5 mm, respectively. Through systematic material screening and laboratory testing, key physical properties of the recycled coarse aggregates, including apparent density, bulk density, water absorption, and crushing index, were determined. The technical specifications are detailed in [Table 2](#).

Table 2. Physical indicators of aggregates.

	Apparent density g/cm ³	Bulk density g/cm ³	Water absorption rate %	Crushing index %	Content of needle-like particles %	Mineral composition %
Coarse aggregate	2.6	1.41	6.5	13.6	8.2	Quartz(75%) Feldspar(12%) Calcite(10%)
Fine aggregate	3.0	1.43	8	-	-	Quartz(88%) Feldspar(10%)

2.1.3. Alkali activators

The NaOH used in this study was produced by Tianjin Ding fu Chemical Plant, supplied in flake solid form with a purity of $\geq 96\%$. The Na₂SiO₃ was a viscous liquid with a modulus of 3.30, containing 8.54% sodium oxide (Na₂O) and 27.3% silicon dioxide (SiO₂). The alkali activator accounted for 14% of the cementitious material's mass. In the NaOH-Na₂SiO₃ composite system, the mass ratio of NaOH to Na₂SiO₃ was 1:2.

2.1.4. Nano-silica

The NS used in this study was sourced from Jinan Zhiding Welding Materials Co., Ltd.

[Table 3](#) presents the basic physical properties of the nano-silica.

Table 3. Physical properties of nano-silica.

Surface properties	Particle size nm	Specific surface area (m ² /g)	Purity (%)	Bulk density (g/cm ³)
Hydrophilic	20	185	99.9	0.16

2.1.5. Fibers

The PPF used in this study were produced by Changsha Hui Xiang Building Materials Co., Ltd. The RCF were sourced from by-products of carbon fiber unidirectional prepreg tapes generated during the production process at Weihai Hao Shi Da Carbon Fiber Technology Co., Ltd. The recycled carbon fiber unidirectional tapes varied in length, with the shortest being approximately 5 mm. Initially, the recycled carbon fibers were screened to separate prepreg materials with uncured surface resin, which were then air-dried to promote resin solidification. The hardened carbon fiber prepreg materials underwent further processing. Large-area carbon fiber prepreg unidirectional tapes were cut perpendicular to the length direction to ensure a maximum length of 40 mm. Subsequently, a crusher was used to process the material, with approximately 30 s of cutting per 100 g of carbon fiber prepreg,

transforming it into RCF [35]. The resulting fibers ranged in length from 5 to 40mm and in diameter from 0.1 to 1 mm. Table 4 presents the performance specifications of the fibers.

Table 4. Fiber performance specifications.

Type of fiber	Density g/cm ³	Length mm	Fiber Diameter um	Tensile strength GPa	Carbon content %
PPF	0.91	19	48	3.0	-
RCF	1.46	5-40	7-10	3.53	95

2.2. Concrete mix design and manufacture

Based on previous studies by Wei et al. in our research group and considering cost-effectiveness, this study employed a fixed water-to-binder ratio of 0.57. The mass dosage of soil-cemented rock (SSR) was set at 9%, 12%, and 15%, also following their findings [7] ... The mass dosage of NS was fixed at 2% of the SSR content [36]. The volume fractions of PPF were 0.10%, 0.30%, and 0.50%, while those of RCF were 0.25%, 0.50%, and 0.75%, all expressed as percentages by volume of concrete. Two alkali-activation systems were employed: a single sodium hydroxide system (denoted as Z) and a composite sodium hydroxide-sodium silicate system (denoted as T). N represents nano-silica (NS). For example, the mix designation NT12-P20-C30 refers to concrete with the composite NaOH-Na₂SiO₃ activation system, incorporating 2% NS relative to SSR, 12% SSR by mass, 0.20% PPF by volume, and 0.30% RCF by volume. The detailed mix proportions for the concrete are presented in Table 5.

Table 5. Mixing ratio and material consumption of each group of SSRRAC.

Number	Content of SSR (%)	Material consumption kg/m ³									
		NaOH	Na ₂ SiO ₃	SSR	Sand	Coarse aggregate	Water	Water reducing agent	NS	Volume content of PPF (%)	Volume content of F (%)
Z12	12	40.18	0	287	873	1004	163	3	0	0	0
NZ12	12	40.18	0	281.26	873	1004	163	3	5.74	0	0
T12	12	13.4	26.8	287	873	1004	163	3	0	0	0
NT12	12	13.4	26.8	281.26	873	1004	163	3	5.74	0	0
T12-P10	15	13.4	26.8	287	873	1004	163	3	0	0.1	0
T12-P30	12	13.4	26.8	287	873	1004	163	3	0	0.3	0
T12-P50	12	13.4	26.8	287	873	1004	163	3	0	0.5	0
T9-P30	9	10.03	20.06	215	924	1038	123	3	0	0.3	0
T15-P30	15	16.75	33.51	359	822	970	204	3	0	0.3	0
T12-C25	12	13.4	26.8	287	873	1004	163	3	0	0	0.25
T12-C50	12	13.4	26.8	287	873	1004	163	3	0	0	0.5
T12-C75	12	13.4	26.8	287	873	1004	163	3	0	0	0.75
T9-C50	9	10.03	20.06	215	924	1038	123	3	0	0	0.5
T15-C50	15	16.75	33.51	359	822	970	204	3	0	0	0.5
NT12-P30	12	13.4	26.8	281.26	873	1004	163	3	5.74	0.3	0
NT12-C50	12	13.4	26.8	281.26	873	1004	163	3	5.74	0	0.5

Number	Content of SSR (%)	Material consumption kg/m ³									
		NaOH	Na ₂ SiO ₃	SSR	Sand	Coarse aggregate	Water	Water reducing agent	NS	Volume content of PPF (%)	Volume content of F (%)
NT12-P20-C30	12	13.4	26.8	281.26	873	1004	163	3	5.74	0.2	0.3

Specimen casting and curing were conducted in accordance with the Standard for Test Methods of Fiber Reinforced Concrete (CECS13-2009) [37]. An HJW-60 single horizontal shaft concrete mixer was used for the preparation and casting of alkali-activated concrete, with the preparation process illustrated in Fig. 1. Initially, recycled aggregates were mixed with saturated water for 1 min to achieve a saturated surface-dry state. Subsequently, RCF were added and mixed for 1 min, utilizing the impact of aggregates to disperse the fibers. Next, sand and SSR were added and mixed for 2 min to ensure uniformity. NS was dispersed in a portion of the water, treated with an ultrasonic cleaner, and then added to the mixer along with the pre-prepared alkali activator solution. The remaining water and superplasticizer were then incorporated, followed by mixing for 2 min. Finally, PPF were added and mixed for 1 min to avoid frictional damage. After cleaning and drying the molds, the concrete mixture was poured, compacted using a vibration machine, and the surface was smoothed to remove excess slurry. The specimens were cured at room temperature for 1 day before demolding and labeling, then maintained at 25°C and 95% relative humidity until the designated curing age.



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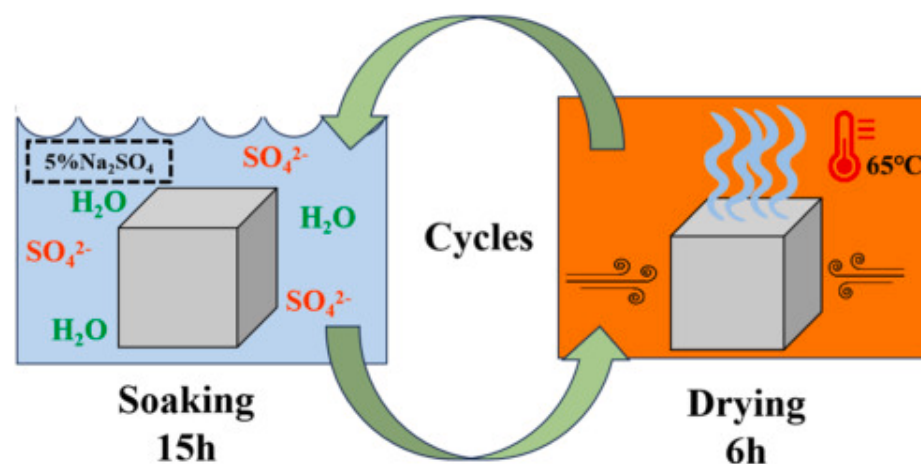
Fig. 1. Schematic diagram of the concrete preparation process.

3. Experimental

3.1. Sulfate attack test

In accordance with the Test Methods of Cement and Concrete for Highway Engineering (JTG 3420-2020) [38], cubic specimens with dimensions of 100mm×100mm×100mm were prepared from the test materials. The specimens were cured in a standard curing

room for 26 days. After curing, the specimens were removed, surface moisture was wiped off, and they were dried in an oven at $65\pm5^{\circ}\text{C}$ for 48h. Following drying, the specimens were cooled to room temperature in a dry environment. Subsequently, the specimens were placed in a concrete sulfate dry-wet cycle testing apparatus, using a 5% sodium sulfate solution as the soaking medium. The solution was periodically refreshed to maintain a stable concentration and chemical environment. The testing apparatus was configured with a soaking period of 15h and a drying period of 6h, as illustrated in Fig. 2. The experiment was conducted strictly according to the specified procedures. This high-temperature dry-wet cycling condition was designed to accelerate sulfate attack, allowing the durability trends of the material to be observed within a limited experimental period. Although the temperature in actual building service is typically lower than 65°C , this condition is suitable for evaluating material durability [39].



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Fig. 2. Schematic diagram of the sulfate test.

The compressive strength test was conducted strictly in accordance with JTG 3420-2020 [38]. Cubic specimens measuring 100mm×100mm×100mm were used, with three parallel specimens tested in each group. A constant loading rate of 0.5–0.8MPa/s was

applied. The arithmetic mean of the three measured values was taken as the compressive strength of the group. If one of the three values deviated from the mean by more than $\pm 10\%$, the average of the remaining two values was adopted. Both the initial compressive strength (after 26d of standard curing) and the compressive strength after specified numbers of dry-wet cycles were determined following the above procedure.

(1) The mass loss rate was evaluated using the formula provided in Eq. (1):

$$K_m = \frac{m_{c0} - m_{cn}}{m_{c0}} \times 100\% \quad (1)$$

Where K_m is the mass loss rate; m_{c0} is the initial mass of the concrete specimen; m_{cn} is the mass of the concrete specimen after n cycles.

(2) The compressive strength corrosion resistance coefficient was calculated using the formula provided in Eq. (2):

$$K_f = \frac{f_{cn}}{f_{c0}} \times 100\% \quad (2)$$

Where K_f is the compressive strength corrosion resistance coefficient (%); f_{cn} is the compressive strength of a group of concrete specimens subjected to sulfate attack after n dry-wet cycles; f_{c0} is the compressive strength of a group of control concrete specimens cured under standard conditions for the same age.

It should be noted that, due to the relatively large surface-to-volume ratio of the 100mm cubic specimens, sulfate attack and ion diffusion are primarily surface-driven, and early-stage internal microcracks may not be immediately reflected in mass change or strength [40]. Therefore, the experimental results are mainly applicable for assessing relative trends in material durability and cannot be directly extrapolated to large-scale structural elements.

3.2. Microstructural analysis

To elucidate the phase composition, chemical bond changes, and microstructural evolution of the cementitious system before and after sulfate dry-wet cycling, XRD, FTIR, and SEM analyses were conducted on concrete specimens. All microstructural samples were obtained from the central regions of fracture fragments from 100mm×100mm×100mm cubic specimens after compressive strength testing.

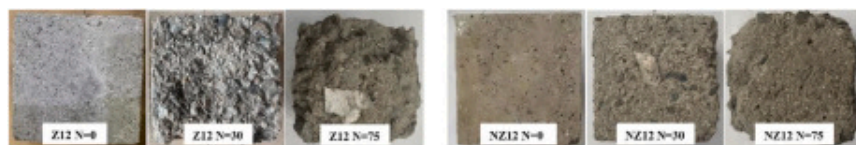
The sample preparation procedure was as follows: immediately after sampling, the fragments were immersed in anhydrous ethanol for 72h to terminate hydration, followed by drying in a vacuum oven at 50°C for 24h. The dried samples were ground using a planetary ball mill and sieved through a 0.080mm sieve, with the undersized powder used for XRD and FTIR analyses. XRD analysis was performed using a Bruker D8 Advance X-ray diffractometer with Cu K α radiation, scanning from 10° to 80° (2 θ) at a step size of 0.02° and a scan rate of 3°/min. FTIR analysis was conducted using an IRTracer-100 Fourier transform infrared spectrometer, scanning from 4000 to 400cm⁻¹ at a resolution of 4cm⁻¹. For SEM, approximately 5mm×5mm×5mm blocks were cut from the fracture fragments and gold-coated under vacuum. SEM observation was performed using a Sigma300 scanning electron microscope.

Concrete specimens, rather than cement paste, were deliberately selected for microstructural analysis primarily for the following reasons: (1) the core objective of this study is to evaluate the sulfate resistance of full solid-waste recycled aggregate concrete under dry-wet cycling, and concrete specimens can more authentically reflect the actual behavior of the ITZ between the binder and recycled aggregates under service conditions; (2) all sulfate attack tests were conducted on concrete specimens, and using the same concrete matrix for microstructural characterization ensures complete consistency between the durability tests and the microscopic investigations.

4. Results and discussion

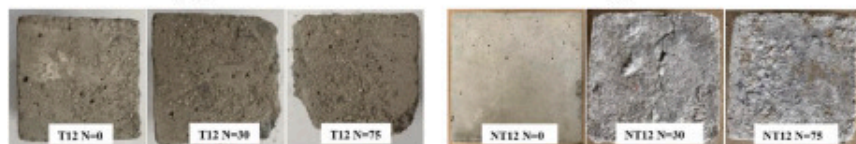
4.1. Surface morphology analysis

To investigate the morphological effects of sulfate attack cycles on mortar specimens (dimensions: 100×100×100mm cubes), tests were conducted with 0, 15, 30, 45, 60, and 75 sulfate attack cycles, documenting the evolution of the specimen surface and structure. Photographs of the surface morphology after 0, 30, and 75 cycles are presented in [Fig. 3](#). Based on the observations, the morphological damage induced by sulfate attack was categorized into five stages, as described below: Stage 1: The specimen surface remains intact with no significant structural damage, but surface roughness slightly increases. Small amounts of white sodium sulfate crystals precipitate, indicating the onset of reactions between sulfate and the mortar matrix. Stage 2: Shallow Edge Fine cracks appear along the edges of the specimen, confined to the surface without penetrating deeper. Sodium sulfate crystal precipitation increases slightly, reflecting stress accumulation in the early stages of erosion. Stage 3: Microcracks expand and interconnect, forming a network of surface cracks. Slight spalling occurs at the edges, with localized material change on the surface, indicating an intensification of damage. Stage 4: Cracks increase significantly in number and penetrate deeper into the specimen, accompanied by noticeable surface swelling. This stage indicates severe structural damage due to expansive stresses caused by sulfate attack. Stage 5: Large-scale spalling occurs at the specimen edges, exposing aggregates on the surface, which are surrounded by substantial white sulfate crystals. The structural integrity is severely compromised. These five stages of surface damage were numbered, and a qualitative assessment of 17 conditions was conducted [[41](#)], as shown in [Fig. 4](#).



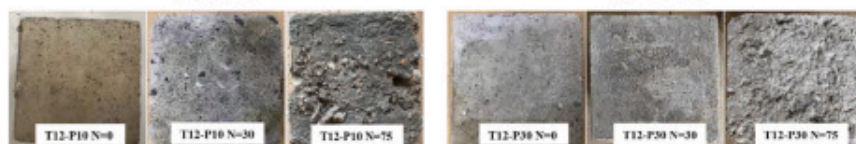
(a) Z12

(b) NZ12



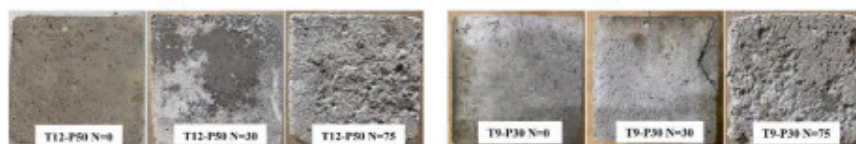
(c) T12

(d) NT12



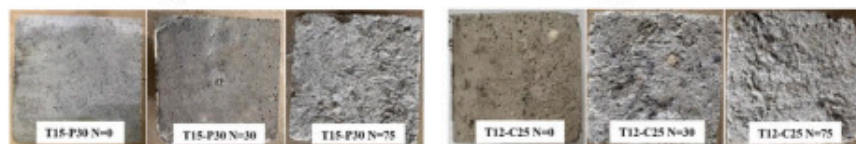
(e) T12-P10

(f) T12-P30



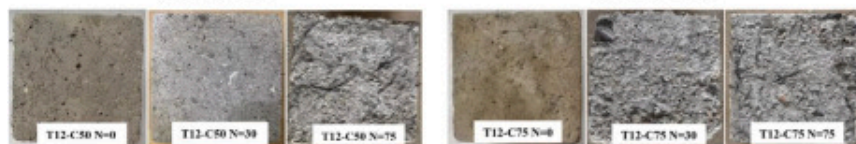
(g) T12-P50

(h) T9-P30



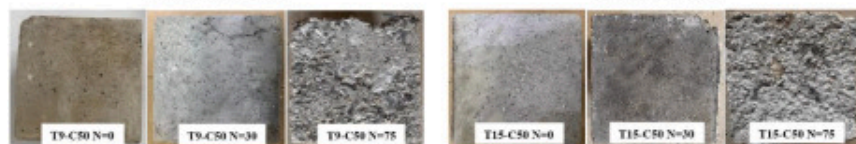
(i) T15-P30

(j) T12-C25



(k) T12-C50

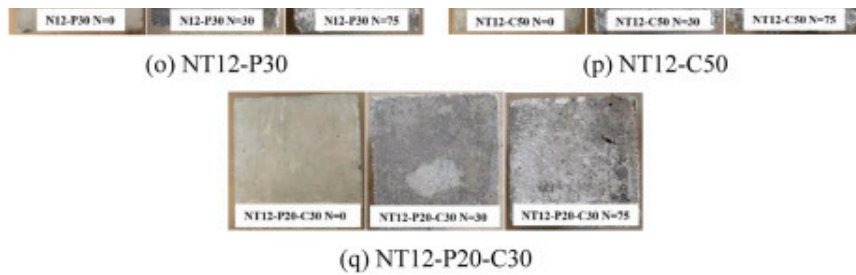
(l) T12-C75



(m) T9-C50

(n) T15-C50

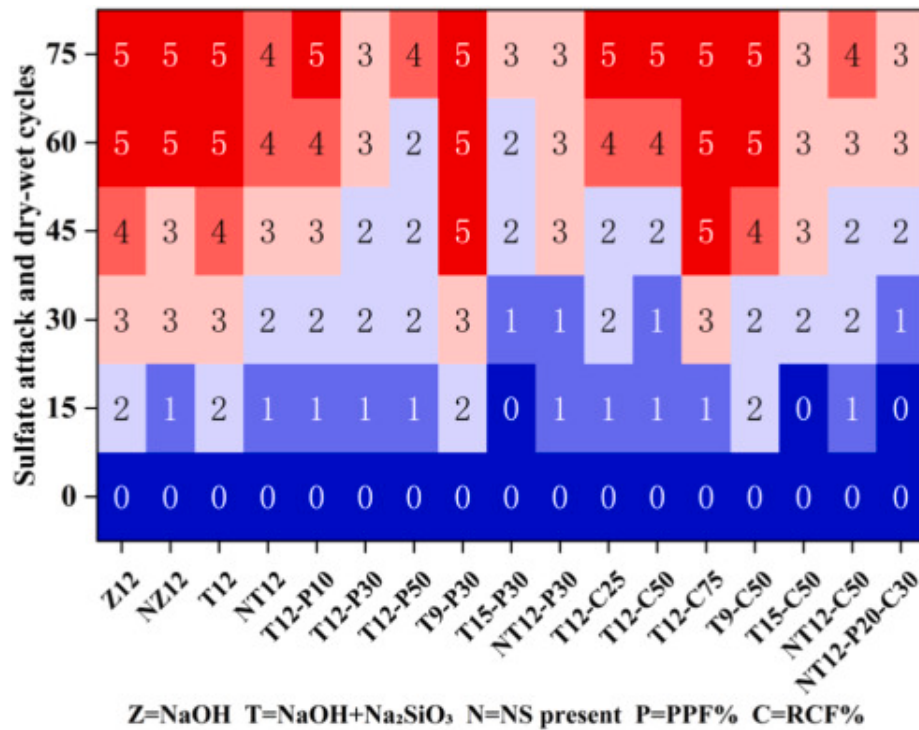




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Fig. 3. Surface morphology of specimens after various cycles of sulfate attack under high-temperature dry-wet conditions.



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Fig. 4. Surface loss assessment of concrete groups.

With an increasing number of sulfate attack and dry-wet cycles, the surface of plain concrete undergoes severe deterioration due to chemical erosion and physical effects, manifesting as surface spalling, cracking, and edge loss. This is attributed to the lack of fiber reinforcement in plain concrete, resulting in low tensile strength and limited resistance to microcrack propagation induced by cyclic stresses. In contrast, fiber-reinforced concrete, benefiting from the bridging effect and reinforcement provided by fibers, effectively suppresses crack development and disperses stresses, exhibiting milder surface damage and superior overall durability.

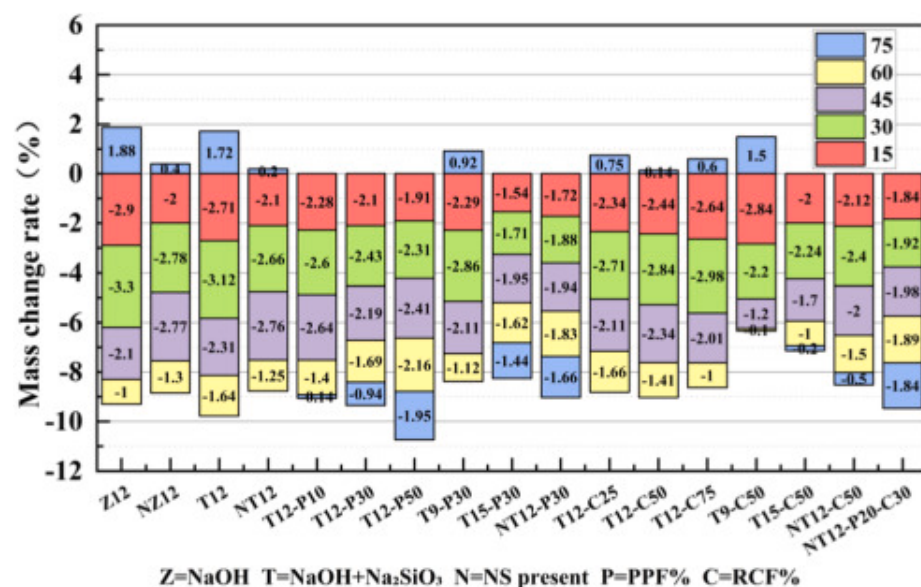
Concrete specimens with 9% SSR content, such as T9-P30 and T9-C50, developed cracks at the edges, as shown in Fig. 3h and m. Other specimens did not exhibit noticeable surface cracks or pores. When the SSR content was increased to 15%, the higher SSR paste content enhanced the concrete's compactness. Additionally, the incorporation of fibers improved the pore structure of SSR, reducing the ingress of sulfate solution and resulting in better structural integrity.

In the early stages of sulfate attack under corrosive environments, the surface and edges of fiber-reinforced concrete specimens remain relatively intact, with only slight sanding observed on the surface structure. This is primarily due to the reaction products of concrete hydration and sulfates gradually filling pre-existing cracks, temporarily enhancing the internal compactness of the specimens [42]. However, under the combined effects of sulfate attack and dry-wet cycles, the SSR matrix experiences partial spalling, resulting in sanding phenomena. As the number of dry-wet cycles increases, the deterioration of fiber-reinforced concrete becomes more pronounced, with edge damage and partial detachment of surrounding aggregates, while the central regions remain relatively intact. Due to sulfate attack, large quantities of corrosion products, such as ettringite and gypsum, form within the internal pore structure. Excessive accumulation of these expansive crystalline products generates internal expansive pressure from the capillary pores, leading to surface cracking and the formation of additional microcracks.

This further facilitates contact between sulfate ions and the internal structure, exacerbating damage.

4.2. Mass change rate analysis

The mass change rates of concrete groups under alternating sulfate attack and dry-wet cycles in a sodium sulfate solution are shown in Fig. 5, where each grid represents the mass change rate of the concrete after a specific cycle. As indicated in Fig. 5, all concrete groups exhibited negative mass growth. However, the incorporation of nano-silica and varying fiber dosages had differing impacts on the mass change rate.



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Fig. 5. Mass change rates of concrete groups under sulfate attack and dry-wet cycles.

After the 15th cycle of sulfate attack under high-temperature dry-wet conditions, plain concrete specimens Z12 and T12 exhibited the highest mass change rates of -2.9% and -2.71%, respectively. In fiber-reinforced concrete, the mass change rates for T9-P30 and

T9-C50 were -2.29% and -2.84% , respectively. In contrast, T15-P30 and T15-C50, with a 15% SSR content, showed significantly lower mass change rates of -1.54% and -2.0% , respectively. When the cementitious material content was 9%, the lower proportion of cementitious material resulted in higher porosity and a looser internal structure, with weaker bonding between the cementitious matrix and aggregates [43]. This significantly reduced the concrete's resistance to sulfate attack. Conversely, increasing the cementitious material content to 15% markedly reduced porosity, enhanced the compactness of the concrete structure, and improved the bonding strength between the cementitious matrix and aggregates, thereby effectively enhancing the concrete's sulfate resistance.

Concrete specimens NZ12 and NT12, incorporating nano-silica, exhibited mass change rates of only -2.0% and -2.1% , respectively. This indicates that in the early stages of sulfate attack, the addition of nano-silica enhances the surface compactness of the concrete. The incorporation of fibers into the matrix optimizes the pore structure, improving specimen density and reducing the ingress of corrosion products and sodium crystals into the concrete interior, thereby significantly mitigating the erosion effect. Additionally, fibers provide a pronounced bridging and stress-transfer effect, effectively restraining the initiation, development, and propagation of cracks, thus alleviating spalling of the matrix. However, the influence of the two fiber types on mass change rate varies with different dosages. Overall, the reduction in mass change rate for PPF reinforced concrete is less pronounced than that for RCF reinforced concrete. This is attributed to the smaller volume and higher quantity of PPF, which effectively connects the matrix and aggregates. As PPF dosage increases, the mass change rate gradually decreases. For instance, the mass change rates of PPF-reinforced concretes T12-P10, T12-P30, and T12-P50 after 75 cycles were -0.14% , -0.94% , and -1.95% , respectively. After 75 cycles of sulfate attack, the mass change rates of concrete with different recycled carbon fiber (RCF) contents showed significant differences: 0.75% for T12-C25, 0.60% for T12-C75, and the lowest value of 0.14% for T12-C50. The optimal performance of T12-C50 is primarily attributed to its moderate fiber content, which allows the fibers to disperse uniformly within the matrix

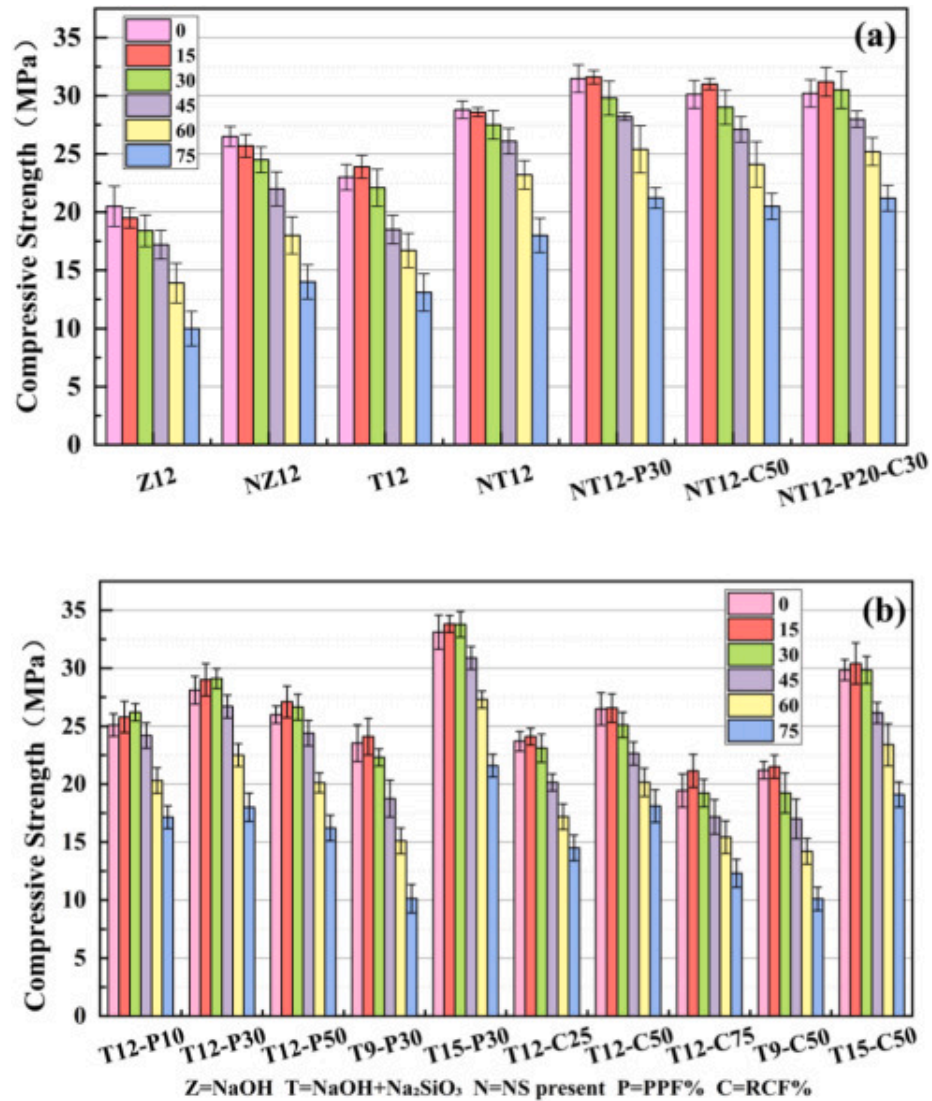
and effectively exert bridging and reinforcing effects. In contrast, the higher fiber content (0.75%) led to fiber agglomeration and increased internal porosity. These defects facilitated the accumulation of expansive products during sulfate attack, resulting in expansion-induced cracking and surface spalling, which explains its relatively higher mass change rate. On the other hand, the lower fiber content (0.25%) provided insufficient fiber quantity and non-uniform dispersion, limiting both the reinforcement of the matrix and the inhibition of crack propagation. Consequently, this mixture exhibited the weakest resistance to sulfate attack and the highest mass change rate among the tested groups. Huang et al. [39] demonstrated that the composite use of high-modulus basalt fibers and high-ductility PPF can achieve mutual support and complementary performance. Under the complex stresses induced by sulfate dry-wet cycles, this combination effectively absorbs stress, exhibiting excellent crack resistance. This finding is consistent with the NT12-P20-C30 system in this study, where RCF, with its high elastic modulus, synergizes with PPF and nanomaterials to achieve the most stable mass change rate under sulfate attack conditions.

Mass change can reflect the overall degradation trend of specimens under sulfate attack. In the early stage, mass may increase due to sulfate ingress and crystallization within the pore structure, whereas subsequent dry-wet cycles lead to expansion, cracking, and surface spalling, resulting in mass loss. Therefore, this indicator is suitable as an auxiliary measure for evaluating macroscopic deterioration. However, for 100mm cubic specimens, early-stage internal microcracks may not produce immediate mass variations, and initial mass gain from crystallization may mask actual damage. Consequently, the sensitivity of this metric is limited, and it cannot independently capture mechanical deterioration; it should be interpreted in conjunction with compressive strength and corrosion-resistance coefficients.

4.3. Compressive Strength Variation

Under high-temperature dry-wet cycles with sulfate attack, the compressive strength of SSR based recycled aggregate concrete specimens modified with nano-silica, PPF, and RCF

varied with the number of cycles, as shown in Fig. 6. As observed in Fig. 6, at (n=15) cycles, the compressive strength of some SSR concrete specimens exhibited an increasing trend. This strength increase can be attributed to two primary factors: first, the concrete continues to hydrate in the corrosive solution [44]; second, corrosion products gradually fill the internal pores of the concrete. In the early stages of erosion, the formation of ettringite initially fills the concrete's pores, enhancing its compactness and thereby increasing compressive strength [45].



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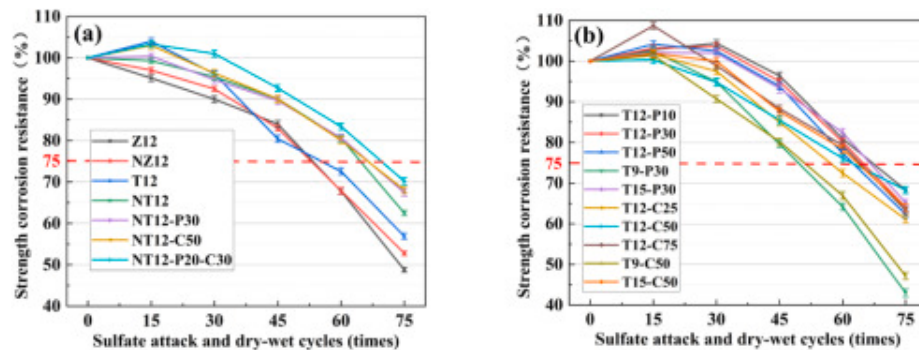
Fig. 6. Compressive strength variation of concrete under sulfate attack: (a) Concrete with different alkali-activation systems and nano-fiber composites (b) concrete with different fiber types and dosages.

As shown in Fig. 6a, in the absence of sulfate attack, the compressive strength of the composite alkali-activation system (NaOH-Na₂SiO₃) was 12.2% higher than that of the single sodium hydroxide (NaOH) alkali-activation system. This improvement is due to the additional silicon source in the composite alkali activator, which results in a denser structure [46,47]. With the addition of nano-silica, the compressive strength of the composite alkali-activation system increased by 25.21%, while that of the single alkali-activation system increased by 29.27%. The enhancement effect of nano-silica is more pronounced in the single alkali-activation system, as water glass (Na₂SiO₃) already provides a high concentration of silicon, limiting the additional effect of nano-silica in the composite system [48].

As shown in Fig. 6b, a comparative analysis of the compressive strength performance of concrete with different fiber dosages at a 28-day curing age was conducted. The compressive strengths of PPF reinforced concretes T12-P10, T12-P30, and T12-P50 were 25.1 MPa, 28.1 MPa, and 26.0 MPa, respectively. For RCF reinforced concretes T12-C25, T12-C50, and T12-C75, the compressive strengths were 23.7 MPa, 26.49 MPa, and 19.44 MPa, respectively. These results indicate that an appropriate fiber dosage significantly enhances the compressive strength of concrete, but excessive fiber content may lead to fiber agglomeration or matrix defects [49], thereby reducing strength. The compressive strength of T12-C75 was the lowest, primarily due to two factors. First, the high fiber content introduced a large amount of air bubbles during mixing [50], reducing the density of the concrete. Second, the recycled carbon fibers (RCF) used in this study were derived from industrial waste, with lengths ranging from approximately 5 to 40 mm and some fibers retaining residual epoxy on the surface, resulting in inherent variability. The residual epoxy caused multiple fibers to stick together and align in parallel, forming microvoids [51,52]. Under high-temperature sulfate wet-dry cycling, these microvoids provided additional pathways for sulfate ion transport, accelerating material degradation. To improve fiber dispersion, hand separation of obvious fiber agglomerates and dry mixing of fibers with aggregates were employed. Experimental results showed that a 0.5% fiber content yielded the optimal mechanical performance, whereas performance

decreased at 0.75%, likely due to fiber agglomeration, uneven length distribution, and residual epoxy creating local weak zones that offset the reinforcing effect.

According to the requirements of the Test Methods of Cement and Concrete for Highway Engineering (JTG 3420-2020) [38], concrete is deemed to have reached the failure criterion when the compressive strength corrosion resistance coefficient drops to 75%. As shown in Fig. 7a, plain concrete specimens Z12, NZ12, and T12 reached this failure threshold after 60 cycles. Similarly, fiber-reinforced concrete specimens T9-P30 and T9-C50 also met the failure criterion after 60 cycles, as depicted in Fig. 7b. After 75 cycles of sulfate attack, the specimens strengthened by the synergistic action of nanomaterials and fibers—NT12-P20-C30, NT12-P30, and NT12-C50—did not reach the “excellent” threshold (75%) defined for conventional concrete. Nevertheless, they exhibited the best performance within the recycled-aggregate concrete system investigated in this study, with compressive strength retention coefficients of 70.2%, 67.43%, and 68.1%, respectively, highlighting the pronounced improvements in sulfate-resistance imparted by the combined use of nanomaterials and fibers.



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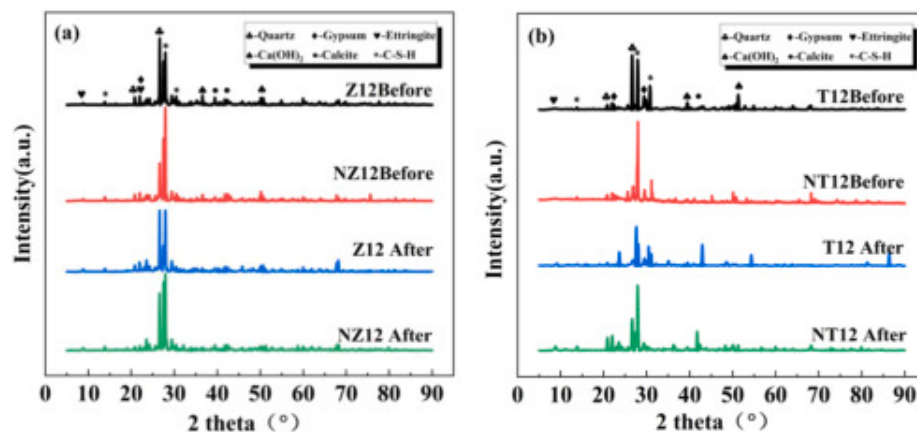
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Fig. 7. Compressive strength corrosion resistance coefficient variation of concrete under sulfate attack: (a) Concrete with different alkali-activation systems and nano-fiber composites (b) concrete with different fiber types and dosages.

4.4. Microscopic investigations

4.4.1. XRD analysis

Fig. 8 presents the XRD patterns of SSR based RAC under single NaOH and composite NaOH–Na₂SiO₃ alkali-activation systems before and after 75 cycles of sulfate attack under high-temperature dry-wet conditions. The patterns reveal the evolution of diffraction peaks for the main phases, including calcium hydroxide (Ca(OH)₂), silicon dioxide (SiO₂), ettringite (Aluminate Ferrite Trisulfate, Aft), calcium carbonate (CaCO₃), C-(A)-S-H, and gypsum (CaSO₄·2H₂O). The positions of the diffraction peaks for each phase remained largely consistent, but their intensities varied depending on the alkali-activation system and the extent of sulfate attack. The quartz peak (SiO₂, 2θ≈26.6°) was the most prominent in all specimens, primarily attributed to the high content of sand and gravel components in the recycled aggregates [53]. Calcium carbonate (CaCO₃, 2θ≈39.41°, 43.1°) was detected in all specimens, likely originating from minor carbonation of raw materials or slight environmental carbonation during testing. The peak intensities of ettringite (Aft, 2θ≈9.09°) and gypsum (CaSO₄·2H₂O, 2θ≈22.52°) reflect the extent of sulfate attack product formation.



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Fig. 8. Xrd patterns of SSR-Based concrete before and after sulfate attack: (a) Single NaOH alkali-activation system (b) composite NaOH–Na₂SiO₃ alkali-activation system.

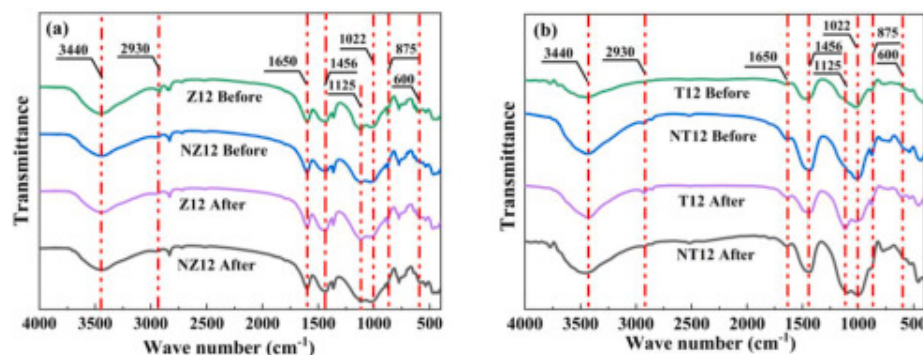
Fig. 8a shows the XRD patterns of Z12 and NZ12 specimens under the single NaOH activation system before and after sulfate attack. Before erosion, both specimens exhibited prominent diffraction peaks for Ca(OH)₂ ($2\theta \approx 34.10^\circ, 51.28^\circ$) and C-(A)-S-H ($2\theta \approx 27.94^\circ$). In comparison, NZ12 displayed an enhanced quartz peak ($2\theta \approx 26.6^\circ$) and a weakened Ca(OH)₂ peak, indicating that the pozzolanic effect of nano-SiO₂ promoted the formation of C-(A)-S-H gel [45]. After erosion, Z12 showed a significant increase in the peak intensities of Aft and gypsum, accompanied by a reduction in Ca(OH)₂ and C-(A)-S-H peaks. This suggests that SO₄²⁻ reacted with Ca(OH)₂ to form CaSO₄·2H₂O, as shown in Eq. (3), followed by the formation of Aft, as shown in Eq. (4). Additionally, sulfate ions underwent an ion exchange reaction with Ca²⁺ in C-(A)-S-H, promoting the migration of Ca²⁺ into the solution, leading to decalcification and decomposition of the C-(A)-S-H structure, as described in Eq. (5). This process exacerbated material deterioration [54]. Severely eroded specimens, particularly Z12, exhibited extensive cracking and loosening of the matrix due to the expansion of Aft and gypsum. During sample grinding, the erosion products were easily pulverized into ultra-fine particles, whereas the hard recycled aggregates remained as relatively coarse fragments. When the powder passing an 80μm sieve was used for XRD analysis, most well-crystallized quartz particles originating from the aggregates were retained on the sieve and excluded from the analyzed sample, resulting in a significant reduction in the intensity of quartz diffraction peaks. In contrast, the increase in Aft peak intensity in NZ12 was less pronounced, indicating that NS partially inhibited sulfate attack. Fig. 8b presents the XRD patterns of T12 and NT12 under the composite NaOH–Na₂SiO₃ activation system. Both specimens showed only weak Ca(OH)₂ diffraction peaks, with high C-(A)-S-H peak intensities, particularly in NT12, where the C-(A)-S-H peak was the most prominent. This suggests that the synergistic effect of Na₂SiO₃ and NS significantly enhanced the formation of dense gel. After erosion, T12 exhibited a slight increase in Aft and CaSO₄·2H₂O peak intensities, while NT12 showed minimal changes, with the C-(A)-S-H peak remaining

nearly unchanged. This indicates superior resistance to sulfate attack under high-temperature dry-wet cycles.



4.4.2. FTIR analysis

Fig. 9 presents the FTIR spectra of SSR based RAC under single NaOH and composite NaOH–Na₂SiO₃ activation systems before and after 75 cycles of sulfate attack under high-temperature dry-wet conditions, revealing changes in chemical bond characteristics. All samples exhibited strong absorption peaks at 3440cm⁻¹ and 1650cm⁻¹, corresponding to the O-H stretching and H-O-H bending vibrations of bound water in Ca(OH)₂ and C-(A)-S-H gel, respectively [55], indicating a complete hydration reaction. Under sulfate attack, the FTIR spectra of the specimens exhibit characteristic absorption bands of SO₄²⁻ at 1100–1130cm⁻¹ and 600–620cm⁻¹, corresponding to the symmetric stretching and bending vibrations, respectively. The marked intensification of these bands confirms the formation of sulfate-bearing corrosion products, primarily ettringite (AFt), gypsum, and monosulfoaluminate [56]. Additionally, the absorption band emerging near 875cm⁻¹, while partly attributable to carbonate species, may also arise from vibrational modes associated with partial Al-for-Si substitution in the C–S–H framework, reflecting localized alterations in the silico-aluminate skeleton following erosion. The asymmetric Si-O-Si/Si-O-Al stretching vibration peak at 1022cm⁻¹ indicated the presence of C-(A)-S-H gel and unreacted SiO₂, corresponding to the vibrations of silica or alumina tetrahedra [57]. This suggests that alkali-activated SSRAC formed a geopolymer structure dominated by Si-O-Si and Si-O-Al chains [58].



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Fig. 9. Ftir spectra of SSR-Based concrete before and after sulfate attack: (a) Single NaOH alkali-activation system (b) composite NaOH–Na₂SiO₃ alkali-activation system.

Fig. 9a illustrates the changes in the NaOH system before and after sulfate attack. Before erosion, the Z12 sample exhibited a strong Si-O-Si/Si-O-Al asymmetric stretching vibration peak at 1022 cm⁻¹. After the addition of NS, the H-O-H peak related to bound water at 1650 cm⁻¹ showed a slight increase in intensity, indicating that the pozzolanic effect of NS promoted the formation of C-(A)-S-H gel, enhancing the matrix's compactness. After sulfate attack, the peak intensity at 1022 cm⁻¹ significantly decreased, reflecting the degradation of the C-(A)-S-H gel structure, consistent with the detection of CaSO₄·2H₂O ($2\theta \approx 22.52^\circ$) in XRD analysis, indicating microstructural deterioration due to sulfate attack. Additionally, the C-H vibration peak at 2930 cm⁻¹ weakened, suggesting the reaction of sulfate ions with Ca(OH)₂ to form sulfate products.

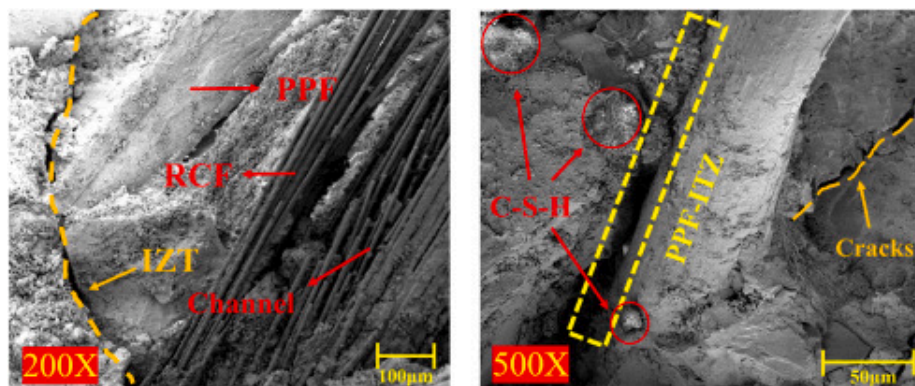
Fig. 9b shows the changes in the NaOH–Na₂SiO₃ composite system before and after sulfate attack, with similar trends in chemical bond changes. After sulfate attack, the C-H vibration peak at 2930 cm⁻¹ significantly decreased or even disappeared, indicating the reaction of sulfate ions with calcium hydroxide. With the addition of 2% nano-silica, the Si-O-related absorption band at 1022 cm⁻¹ showed no significant change, suggesting that nano-silica does not alter the gel structure of SSRRAC in the NaOH–Na₂SiO₃ system [59].

This may be attributed to the sufficient silicon source provided by sodium silicate in the composite system, which overshadows the contribution of nano-silica. In the NaOH–Na₂SiO₃ composite activation system, the characteristic CaCO₃ absorption peak at 1456 cm⁻¹ is markedly higher than in NaOH-only activated specimens. This can be attributed to the elevated alkalinity imparted by Na₂SiO₃, which enhances carbonation reactions with atmospheric CO₂ during specimen preparation and curing. Moreover, the denser C-(A)-S-H gel formed in the composite system provides additional nucleation and adhesion sites for CaCO₃. Under the conditions of this study, the extent of carbonation remains moderate, with no observable adverse effects on overall compressive strength or interfacial zone density.

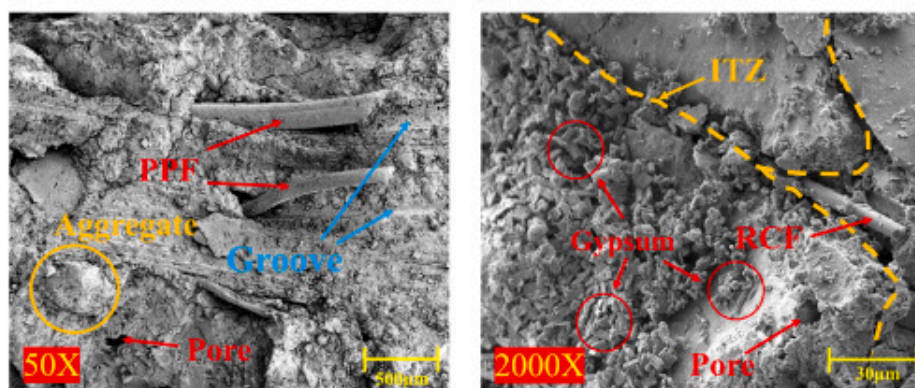
4.4.3. SEM analysis

Fig. 10 presents SEM images of SSRRAC (NT12-P20-C30) subjected to varying cycles of sulfate attack under high-temperature dry-wet conditions. As shown in Fig. 10a, for concrete not exposed to sulfate attack, the SSR matrix surrounding the fibers appears dense. The incorporation of fibers provides a “bridging” effect, controlling crack propagation. This indicates that the bond between fibers and the concrete matrix is critical for the fibers to effectively serve as bridges [60]. The addition of nano-silica enhances the internal compactness of the concrete, reducing the ingress of sulfate ions. Additionally, fibers bear the pressure from crystal expansion within the concrete, making the combination of fibers and nano-silica superior in sulfate resistance compared to other groups. After 30 cycles of sulfate attack, prismatic crystals with lengths of 10–20 μm, identified as gypsum, were observed in the concrete, as shown in Fig. 10b. These crystals cause expansive cracking of the matrix, reducing the bonding strength of the C-(A)-S-H gel and leading to a decrease in specimen strength. As depicted in Fig. 10b, the mortar matrix contains pores and grooves, with fiber surfaces showing signs of roughness due to the pull-out process from the matrix. After 75 cycles of sulfate attack, the surface concrete exhibits severe deterioration, with the SSR matrix gradually spalling. However, the interlaced fibers on the concrete surface mitigate the spalling of the mortar, as shown in Fig. 10c [61]. Despite the presence of numerous macrocracks and pores in the mortar

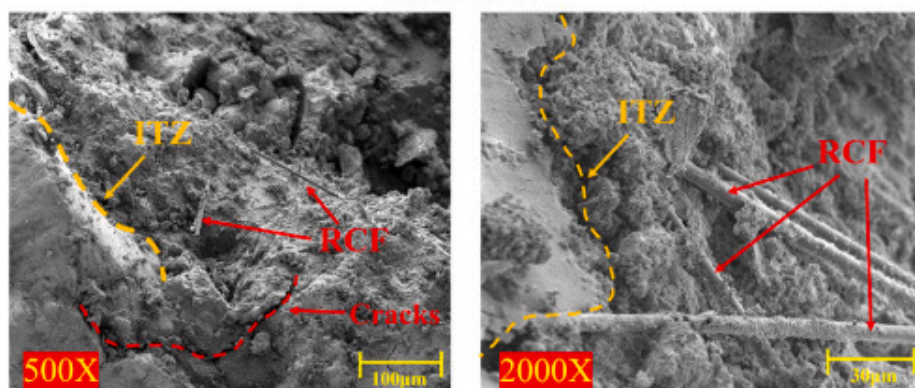
matrix, the fibers maintain their bridging state, attributed to the strong bond between the fibers and the mortar matrix, as well as the high elastic modulus of the carbon fibers.



(a) 0 Cycles



(b) 30 Cycles



(c) 75 Cycles

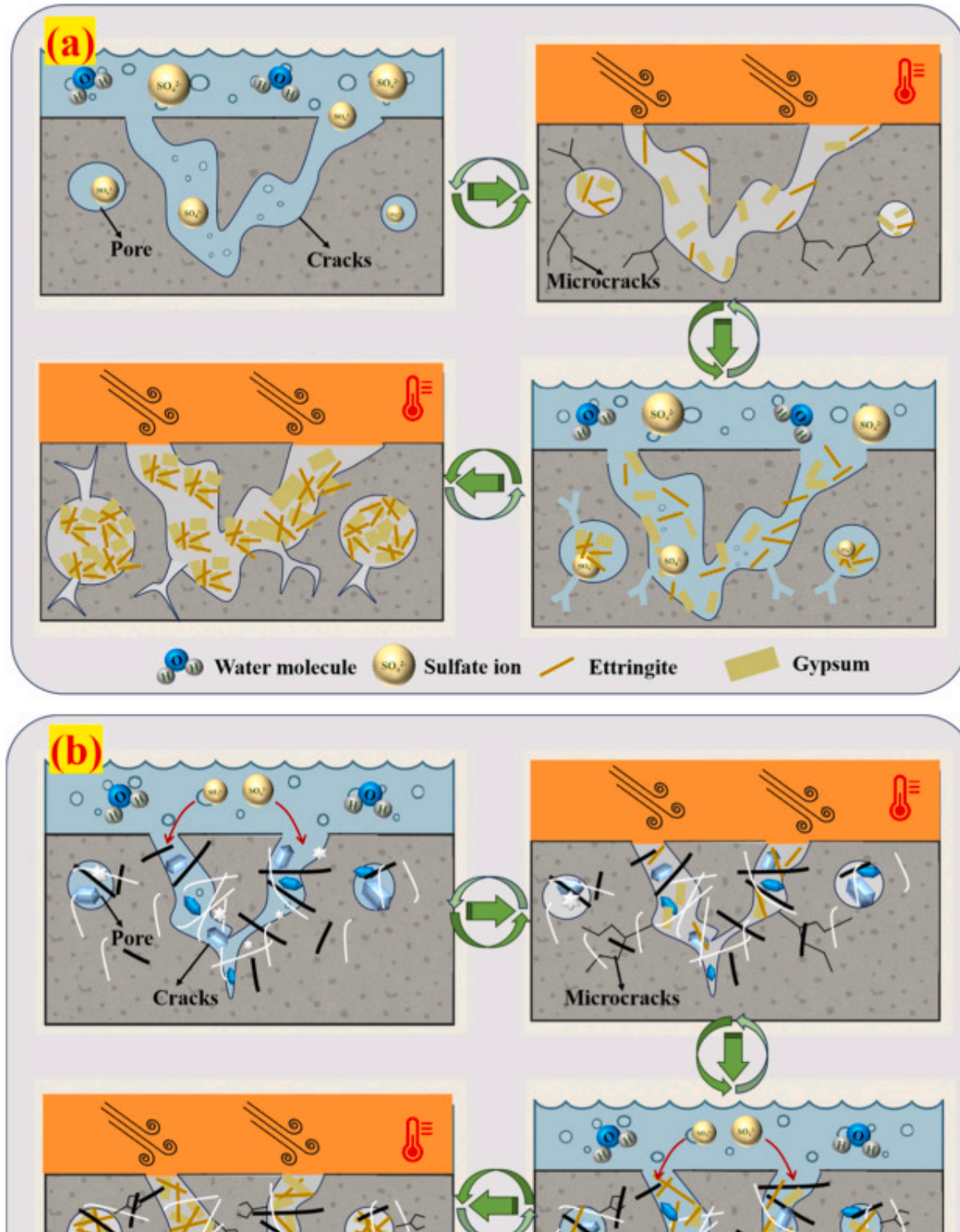
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Fig. 10. Sem images of concrete after sulfate attack: (a) 0 cycles; (b) 30 cycles; (c) 75 cycles.

4.5. Mechanism analysis of nano-fiber synergistic enhancement in concrete under sulfate attack

Fig. 11 illustrates the schematic diagram of the degradation mechanisms for plain concrete and nano-fiber composite modified concrete under sulfate attack. The sulfate attack process exhibits a dynamic cyclic pattern characterized by “filling–degradation–refilling–redegradation.” In the initial stage, corrosion products gradually fill the fine pores within the concrete, reducing pore size. As these products come into contact with the pore walls, localized expansive stresses are generated due to crystal expansion. When these stresses exceed the tensile strength of the matrix, pore expansion and rupture occur, forming larger pores and secondary microcracks, thereby releasing the crystal expansion energy. Subsequently, new sulfate corrosion products further fill these expanded pores, temporarily increasing the compactness of the pore structure until the expansive pressure accumulates again, triggering another cycle of pore degradation [62], as shown in Fig. 11a.





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Fig. 11. Schematic diagram of sulfate attack reaction mechanisms: (a) Plain concrete; (b) nano-fiber reinforced concrete.

In the nano-fiber composite system shown in Fig. 11b, the synergistic action of fibers and nanomaterials significantly enhances the corrosion resistance of concrete. The incorporation of fibers forms a randomly distributed three-dimensional network structure within the concrete, which refines the pore structure through a filling effect, shortening the diffusion paths of corrosive ions. Additionally, the effective interfacial bonding between fibers and the SSR matrix resists tensile and shear stresses induced by the formation of corrosion products, thereby suppressing microcrack propagation [62]. Meanwhile, the ultra-fine particle size and high reactivity of NS promote secondary hydration reactions with Ca(OH)_2 in the alkali-activation system, generating a higher quantity of dense C-(A)-S-H gel, which enhances the matrix's compactness and mechanical strength. However, during sulfate attack, this C-(A)-S-H gel remains susceptible to calcium ion loss and structural degradation, exhibiting an evolutionary process of “initial enhancement followed by decomposition.” Nevertheless, the combination of C-(A)-S-H gel formation promoted by NS and the physical reinforcement provided by fibers effectively delays the accumulation of corrosion products and the rate of crack propagation, achieving the goal of synergistically enhancing sulfate resistance.

5. Conclusions

This study utilized SSR as the cementitious material and 100% recycled coarse aggregates to prepare alkali-activated RAC. Through sulfate attack tests, compressive strength tests, and characterization techniques including XRD, FTIR, and SEM, the effects of incorporating PPF, RCF, and NS on the durability under high-temperature dry-wet sulfate attack cycles and the underlying microscopic mechanisms were systematically investigated. The following conclusions were drawn.

- (1) The incorporation of polypropylene fibers (PPF) and recycled carbon fibers (RCF) significantly enhanced both the compressive strength and sulfate resistance of recycled aggregate concrete (RAC). Among them, 0.3% PPF (T12-P30) exhibited the most pronounced improvement in compressive strength, achieving a 28-day compressive strength of 28.1 MPa. In contrast, 0.1% PPF (T12-P10) demonstrated the best resistance to sulfate attack, with a mass change of only -0.14% after 75 cycles. By comparison, 0.5% RCF (T12-C50) achieved a well-balanced performance between mechanical strength and durability, yielding a 28-day compressive strength of 26.49 MPa and a mass change rate of 0.14% after 75 cycles. These findings indicate that the flexible bridging effect of PPF is particularly effective in suppressing microcrack propagation and mitigating erosion damage, whereas the high modulus of RCF contributes more directly to strength enhancement. This study provides clear and valuable guidance for the optimal selection of fiber type and dosage in fiber-reinforced RAC.
- (2) The incorporation of NS enhanced the compressive strength of concrete under the single NaOH activation system by 29.27%, outperforming the NaOH-Na₂SiO₃ composite system, likely due to the additional silicon source in the composite system limiting the contribution of NS. The synergistic modification with NS, PPF, and RCF significantly improved the performance of alkali-activated RAC. The NT12-P20-C30 specimen achieved a compressive strength of 30.2 MPa after 28 days of curing, representing a 31.3% increase compared to plain T12 concrete. After 75 days of sulfate

attack, NT12-P20-C30 exhibited a mass change rate of only -1.84% and a compressive strength corrosion resistance coefficient of 70.2%, significantly surpassing other groups and demonstrating exceptional corrosion resistance.

- (3) XRD and FTIR analyses revealed that NS promoted secondary hydration reactions through its pozzolanic effect, significantly reducing Ca(OH)₂ crystal content and generating more C-(A)-S-H gel. This optimized the microstructure of the ITZ and reduced porosity. In sulfate attack environments, NS-modified specimens exhibited significantly lower peak intensities for Aft and gypsum, indicating that NS effectively impeded the penetration of corrosive substances and suppressed the formation of expansive products, thereby enhancing sulfate resistance.
- (4) SEM analysis showed that hydration products adhered to fiber surfaces, strengthening the fiber-matrix interfacial bonding and effectively suppressing the initiation and propagation of microcracks. NS further enhanced matrix compactness by filling micropores and optimizing the ITZ structure. The bridging effect of fibers combined with the filling effect of NS formed a three-dimensional network structure, shortening the diffusion paths of corrosive ions, resisting expansive stresses, and significantly improving the pore structure, mechanical properties, and long-term durability of RAC.
- (5) Sulfate attack exhibits a dynamic cyclic pattern of “filling–degradation–refilling–redegradation.” In the initial stage, corrosion products fill micropores, while in later stages, expansive stresses lead to pore expansion and microcrack formation. The synergistic effect of NS and fibers delays the accumulation of corrosion products and the rate of crack propagation by forming dense gels, refining the pore structure, and enhancing interfacial

bonding strength. This provides reliable assurance for the application of RAC in harsh environments.

Future research could employ advanced techniques such as synchrotron XRD, NMR, X-ray computed tomography (X-CT), and mercury intrusion porosimetry (MIP) to investigate the hydration reaction pathways and C-(A)-S-H gel formation mechanisms of SSR under NaOH and NaOH–Na₂SiO₃ systems, enabling the development of kinetic models. Additionally, monitoring the evolution of pore structures in RAC during sulfate attack and elucidating their correlation with compressive strength and corrosion resistance could facilitate the establishment of microscopic damage prediction models, providing a theoretical basis for optimizing alkali-activation systems and assessing long-term durability.

CRedit authorship contribution statement

Haifeng Wan: Funding acquisition. **Guangqiang Liu:** Writing – original draft. **Lei Gao:** Project administration. **Zuowei Liu:** Funding acquisition. **Shuo Yuan:** Methodology. **Huijie Wei:** Data curation. **Chao Dong:** Project administration. **Zhaodi Yuan:** Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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